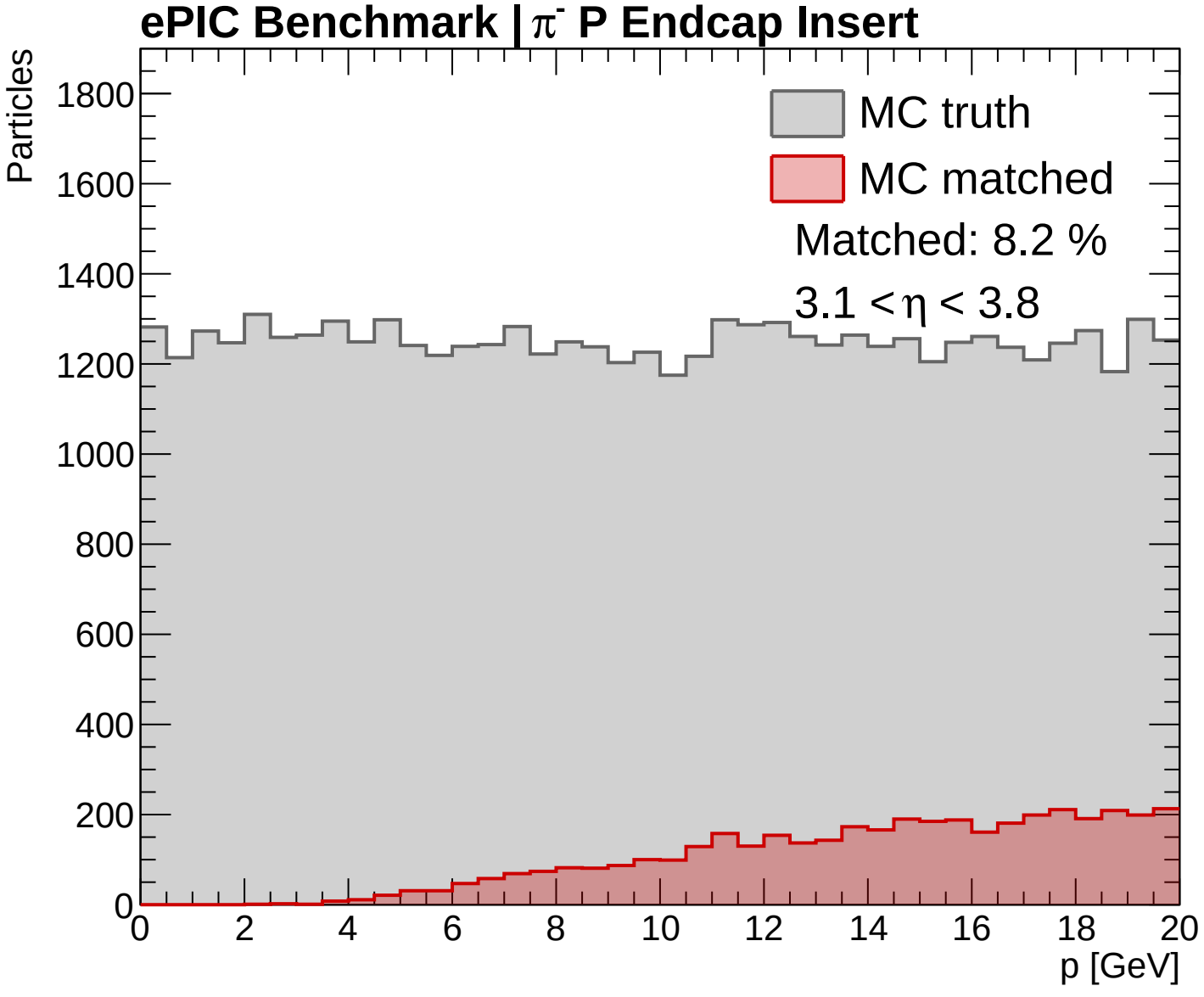
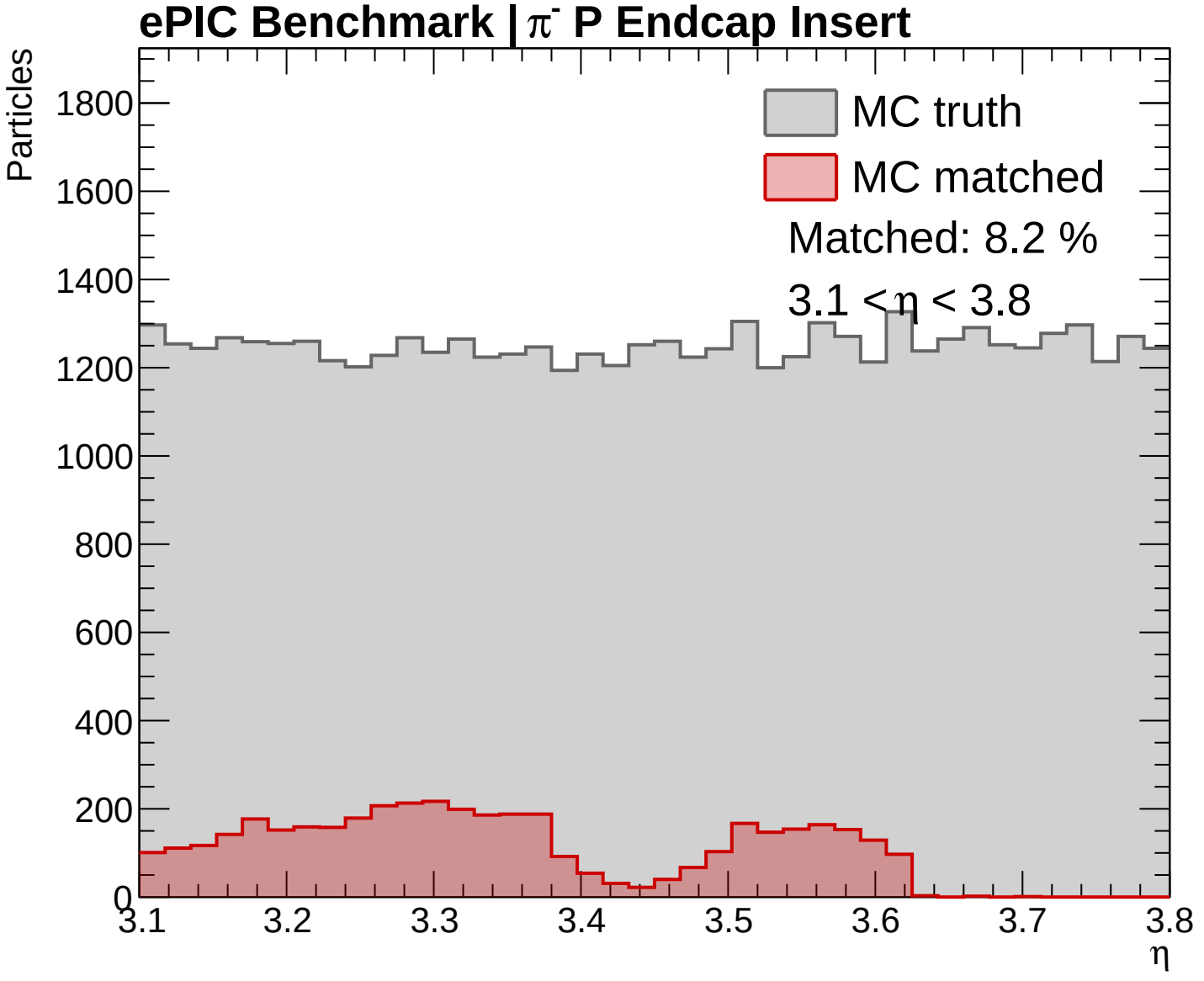
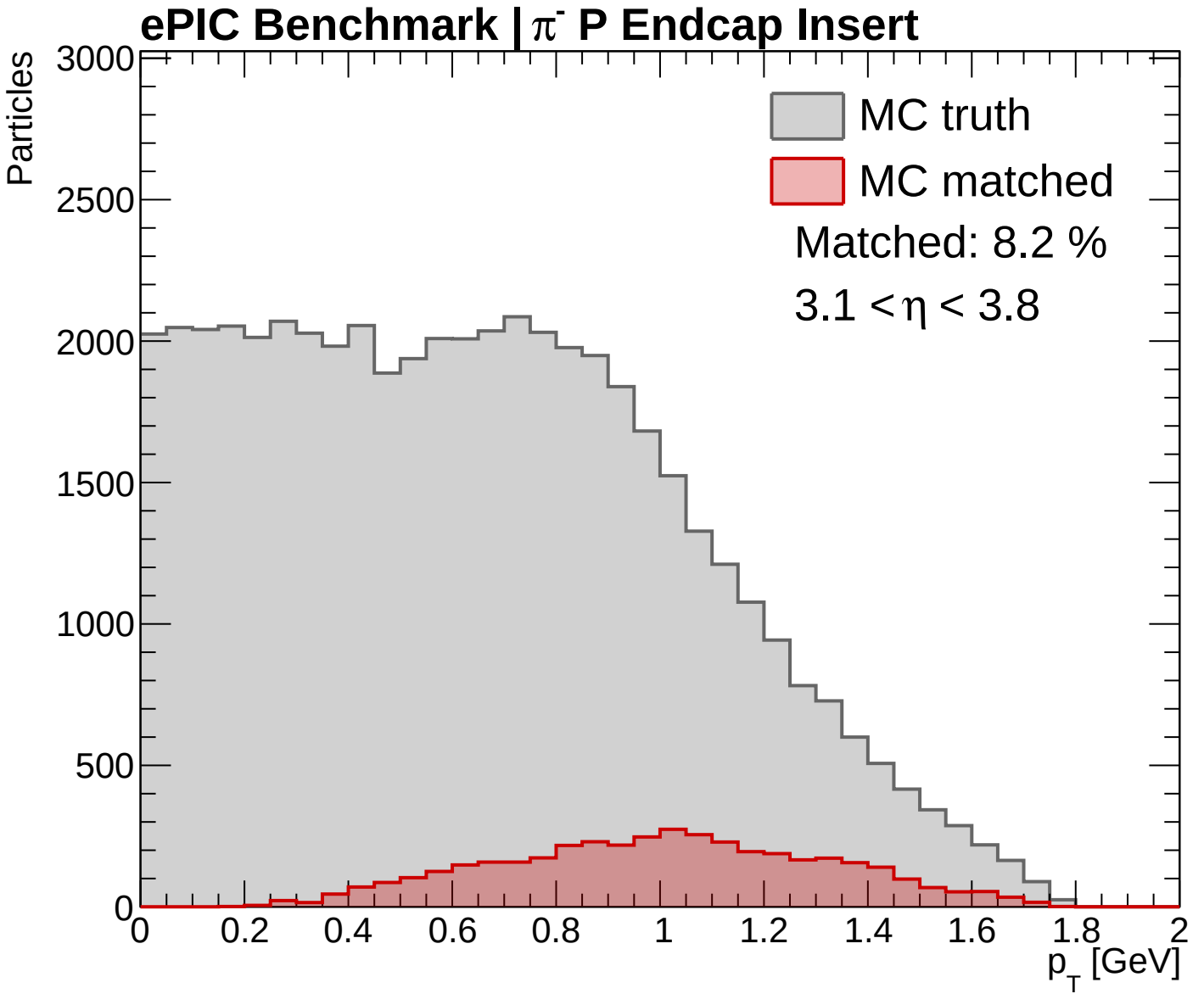
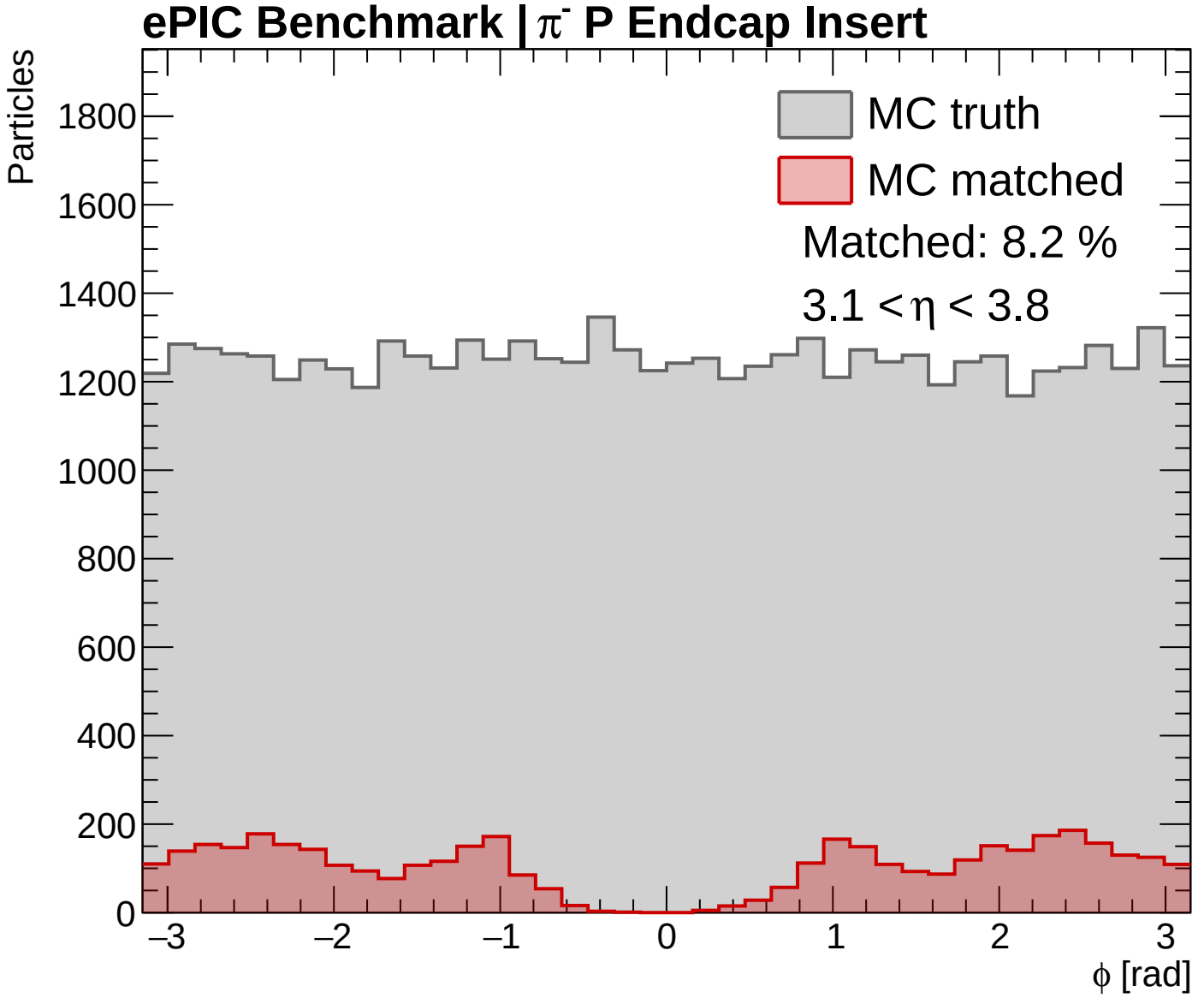
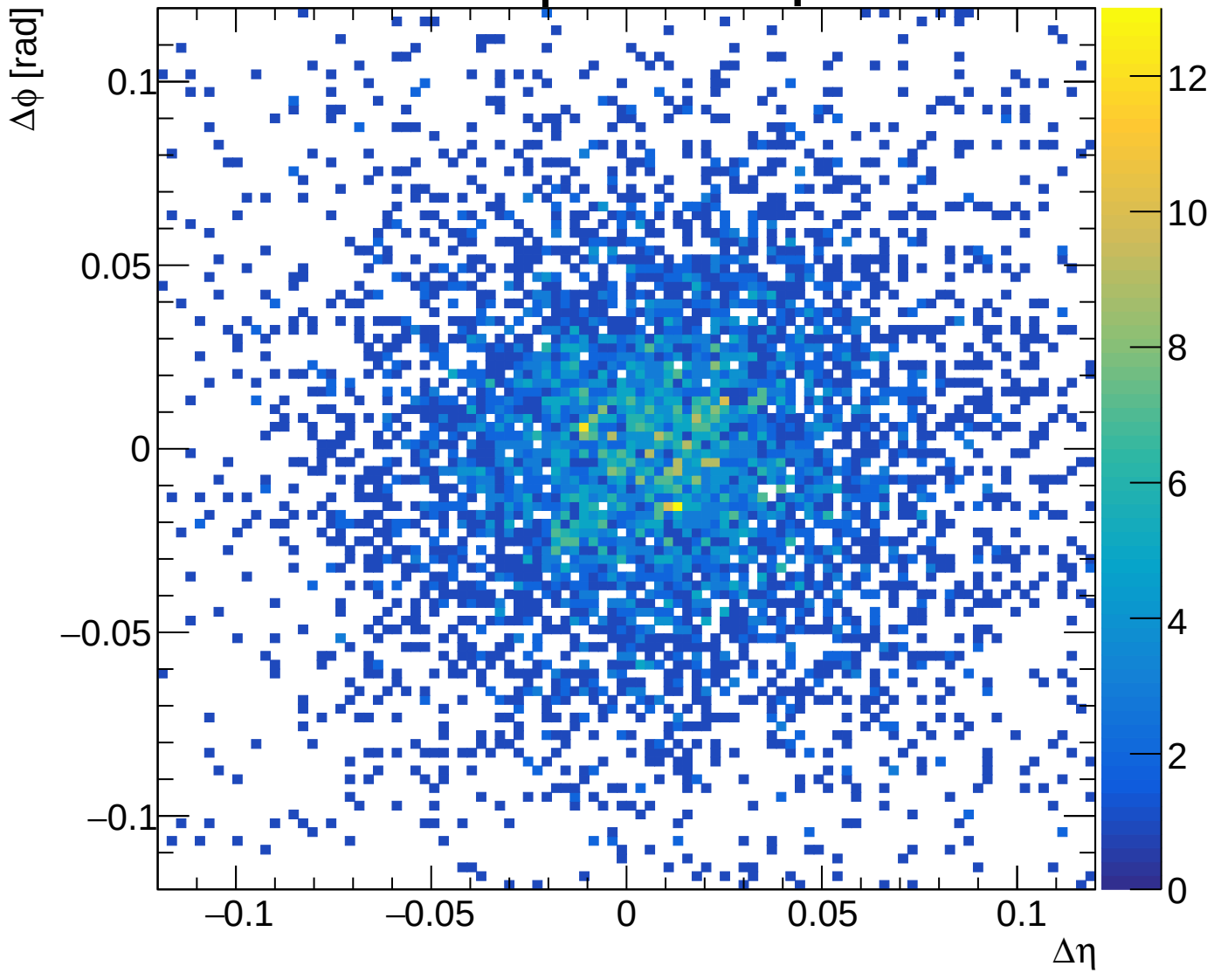
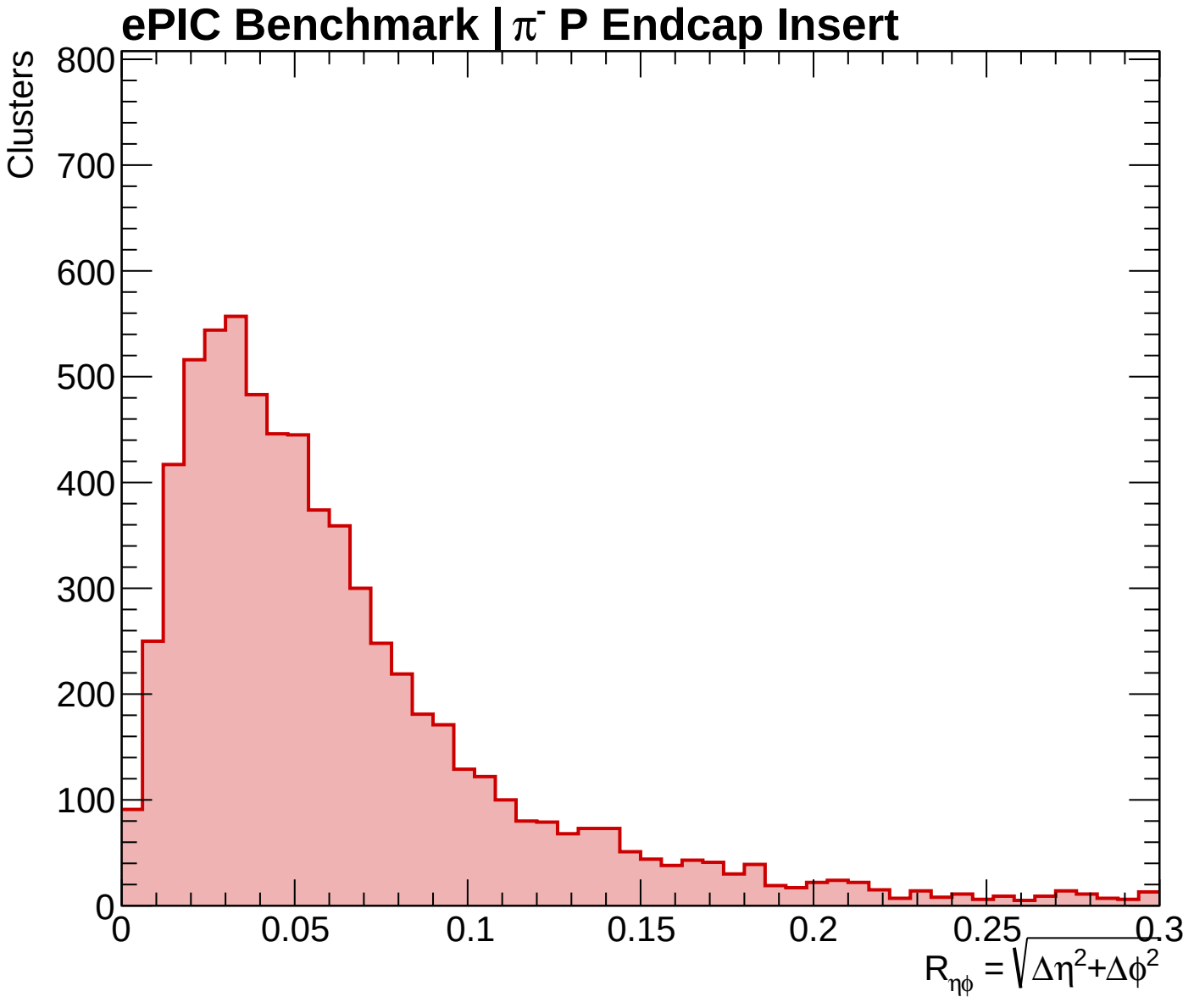

ePIC Benchmark | π⁻ P Endcap Insert
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
0
2
4
6
8
10
12
14
16
18
20
p [GeV]
MC truth
MC matched
Matched: 8.2 %
3.1 < η < 3.8

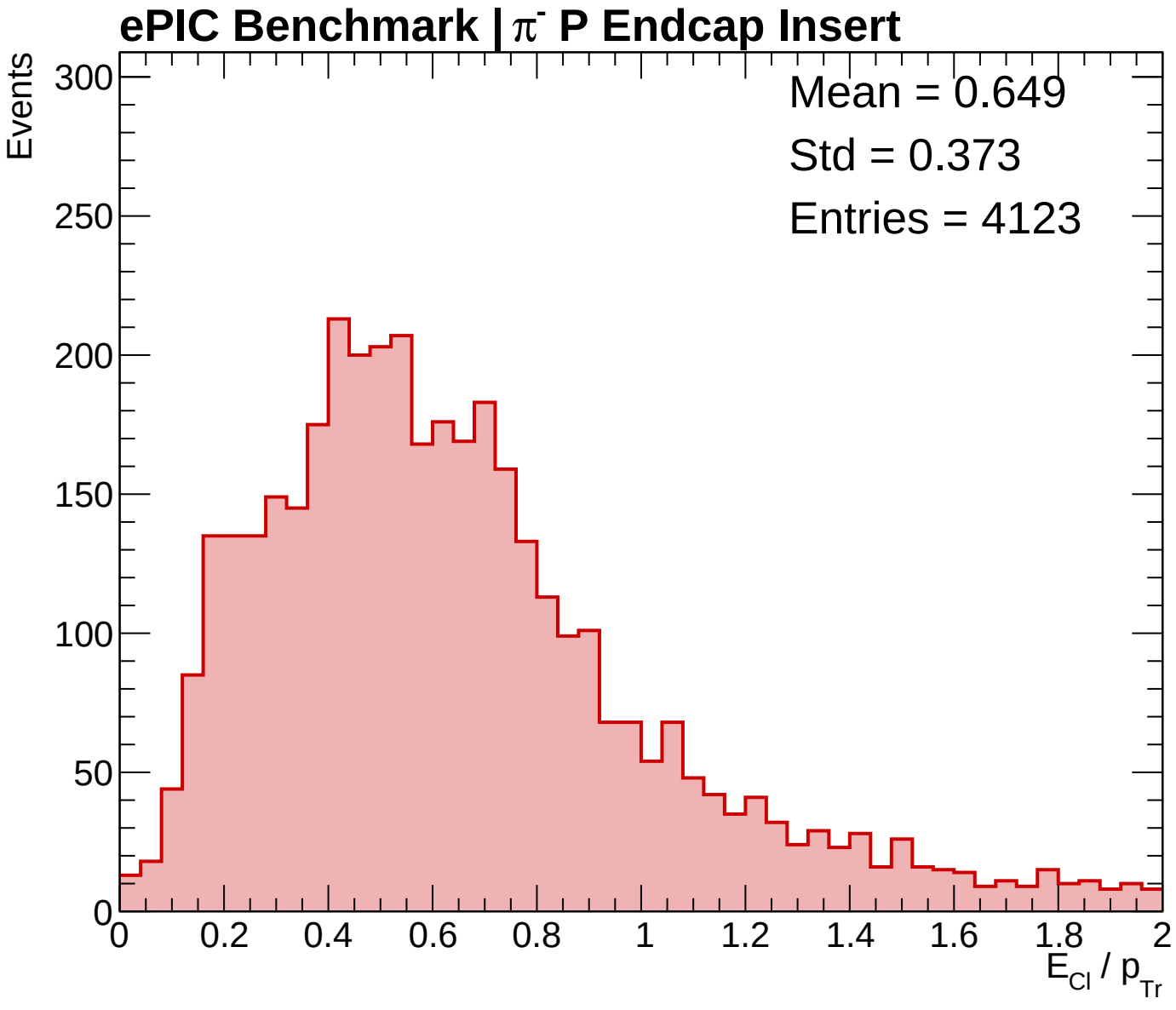

ePIC Benchmark | π⁻ P Endcap Insert
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
3.1
3.2
3.3
3.4
3.5
3.6
3.7
3.8
η
MC truth
MC matched
Matched: 8.2 %
3.1 < η < 3.8

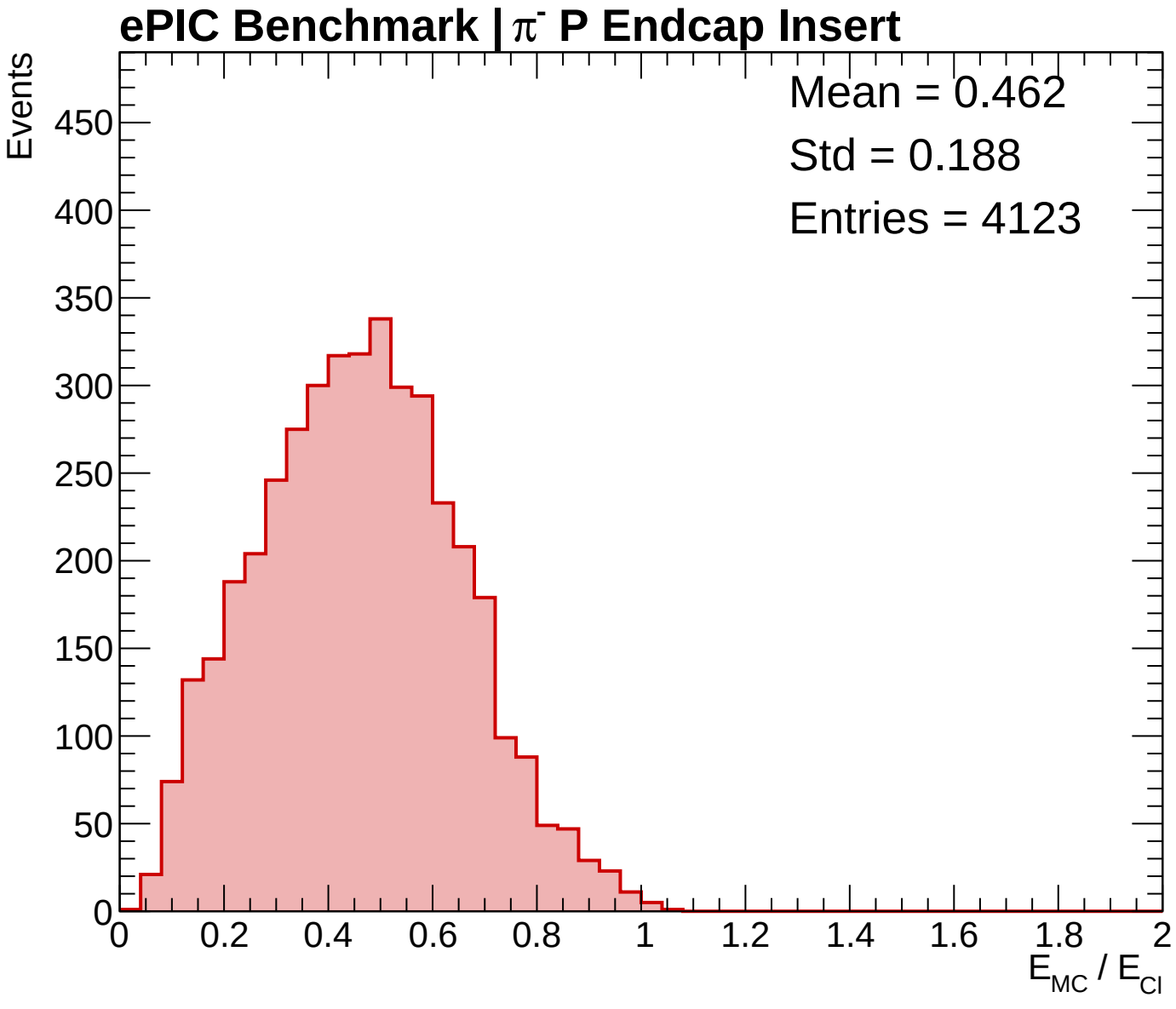

ePIC Benchmark | π⁻ P Endcap Insert
Particles
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
p_T [GeV]
MC truth
MC matched
Matched: 8.2 %
3.1 < η < 3.8

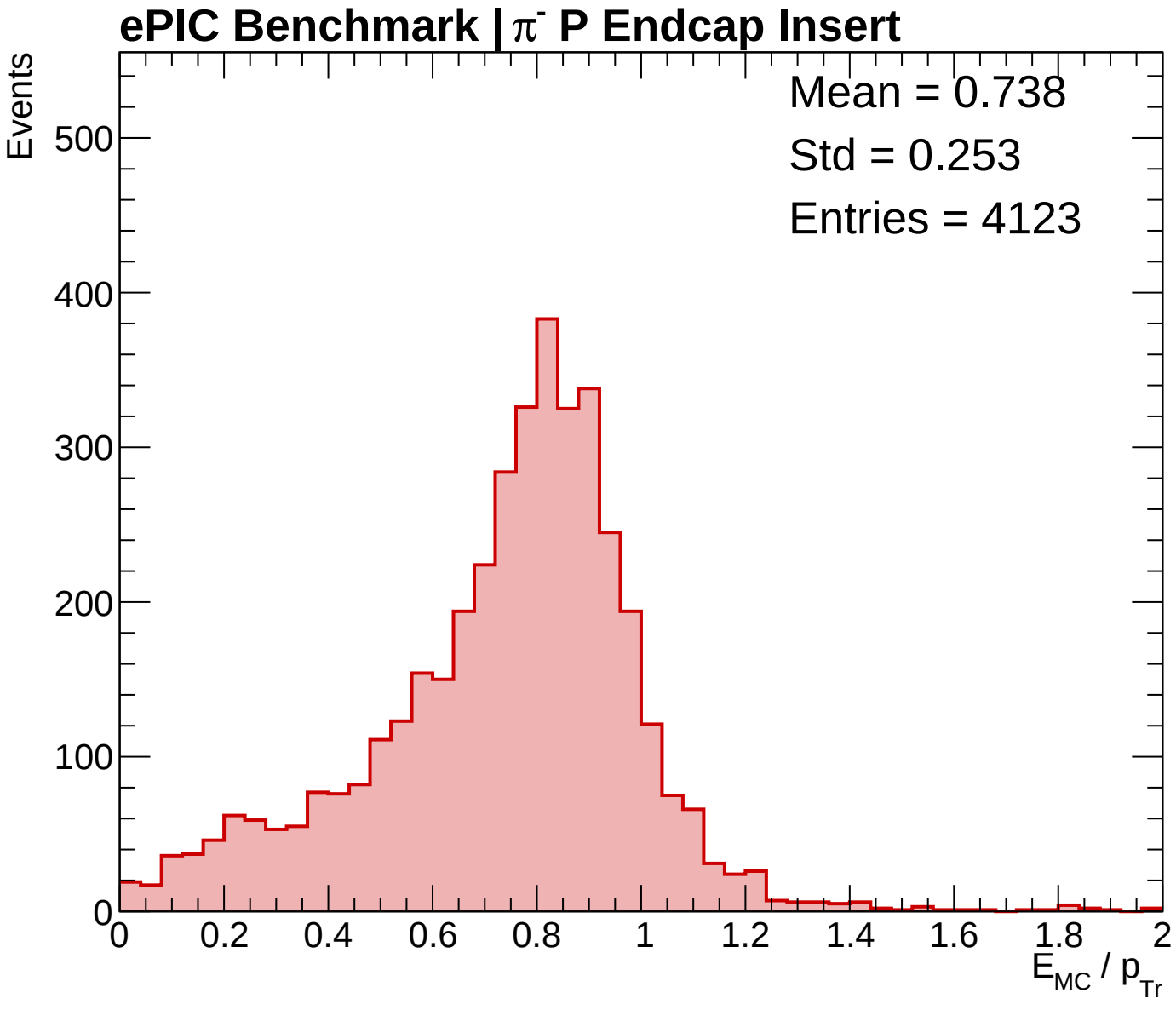

ePIC Benchmark | π⁻ P Endcap Insert
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
–3
–2
–1
0
1
2
3
ϕ [rad]
MC truth
MC matched
Matched: 8.2 %
3.1 < η < 3.8

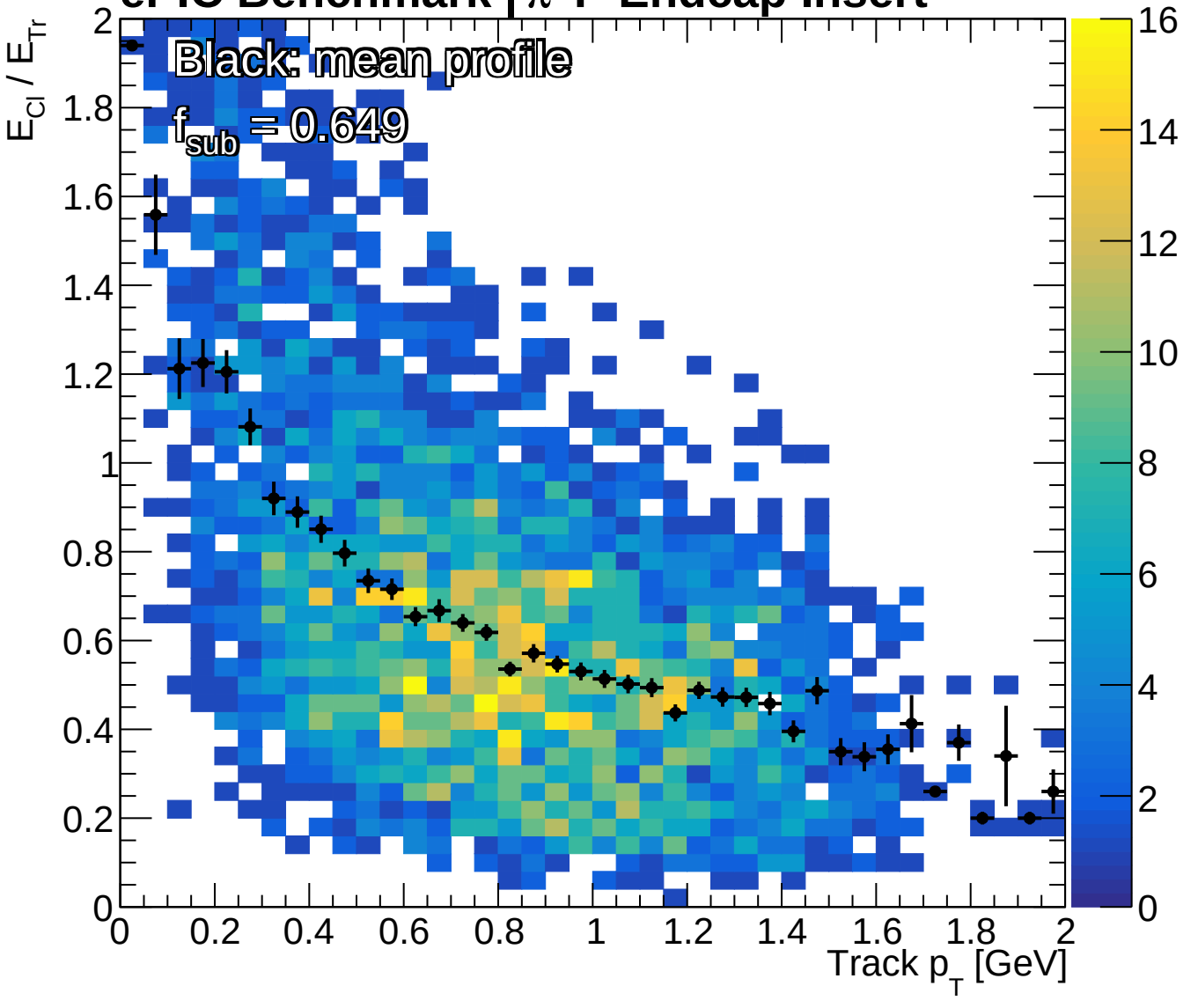

ePIC Benchmark | π⁻ P Endcap Insert
Δϕ [rad]
0.1
0.05
0
−0.05
−0.1
−0.1
−0.05
0
0.05
0.1
Δη
12
10
8
6
4
2
0

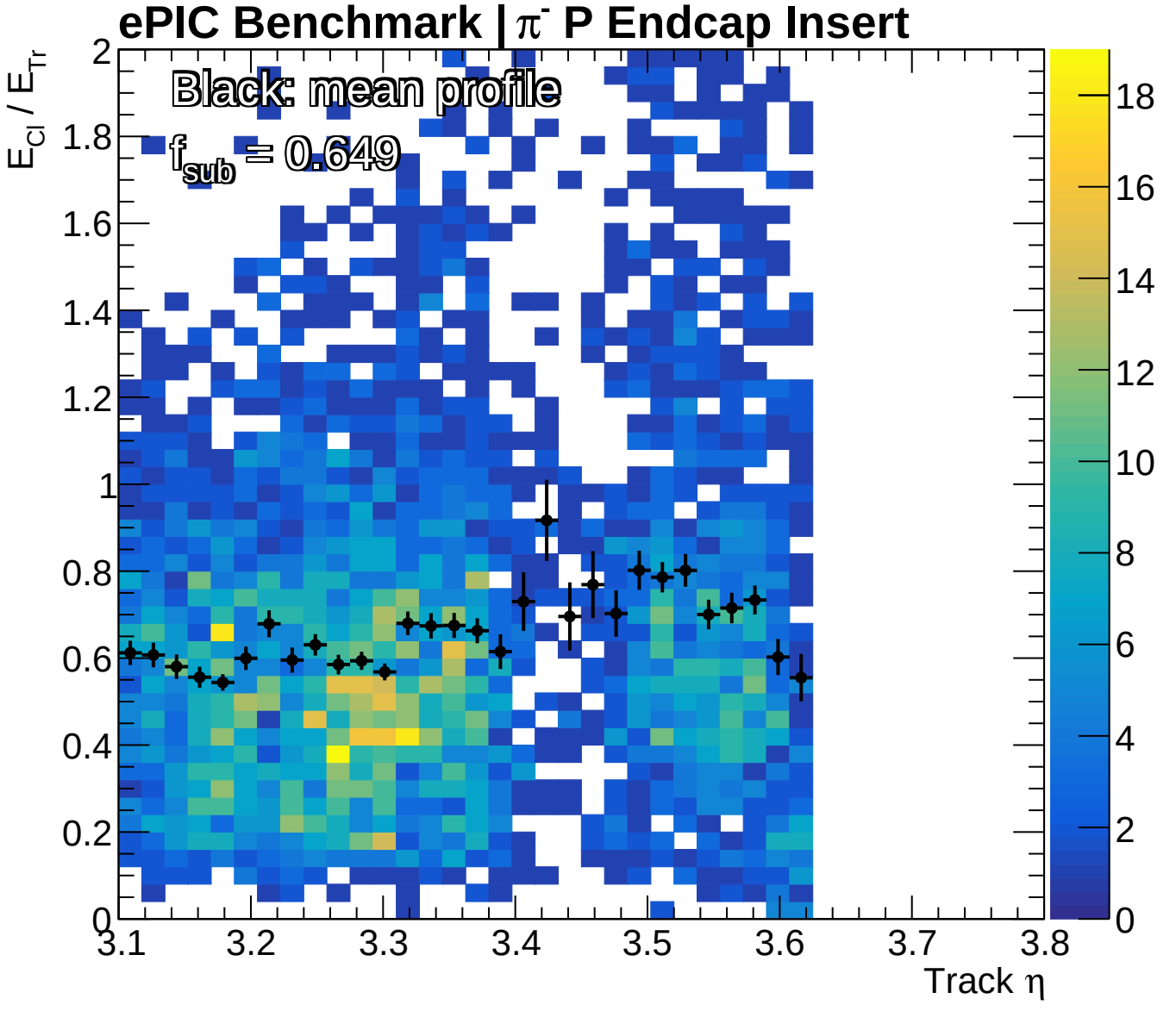

ePIC Benchmark | π⁻ P Endcap Insert
Clusters
800
700
600
500
400
300
200
100
0
0
0.05
0.1
0.15
0.2
0.25
0.3
$R_{\eta\phi} = \sqrt{\Delta\eta^2+\Delta\phi^2}$

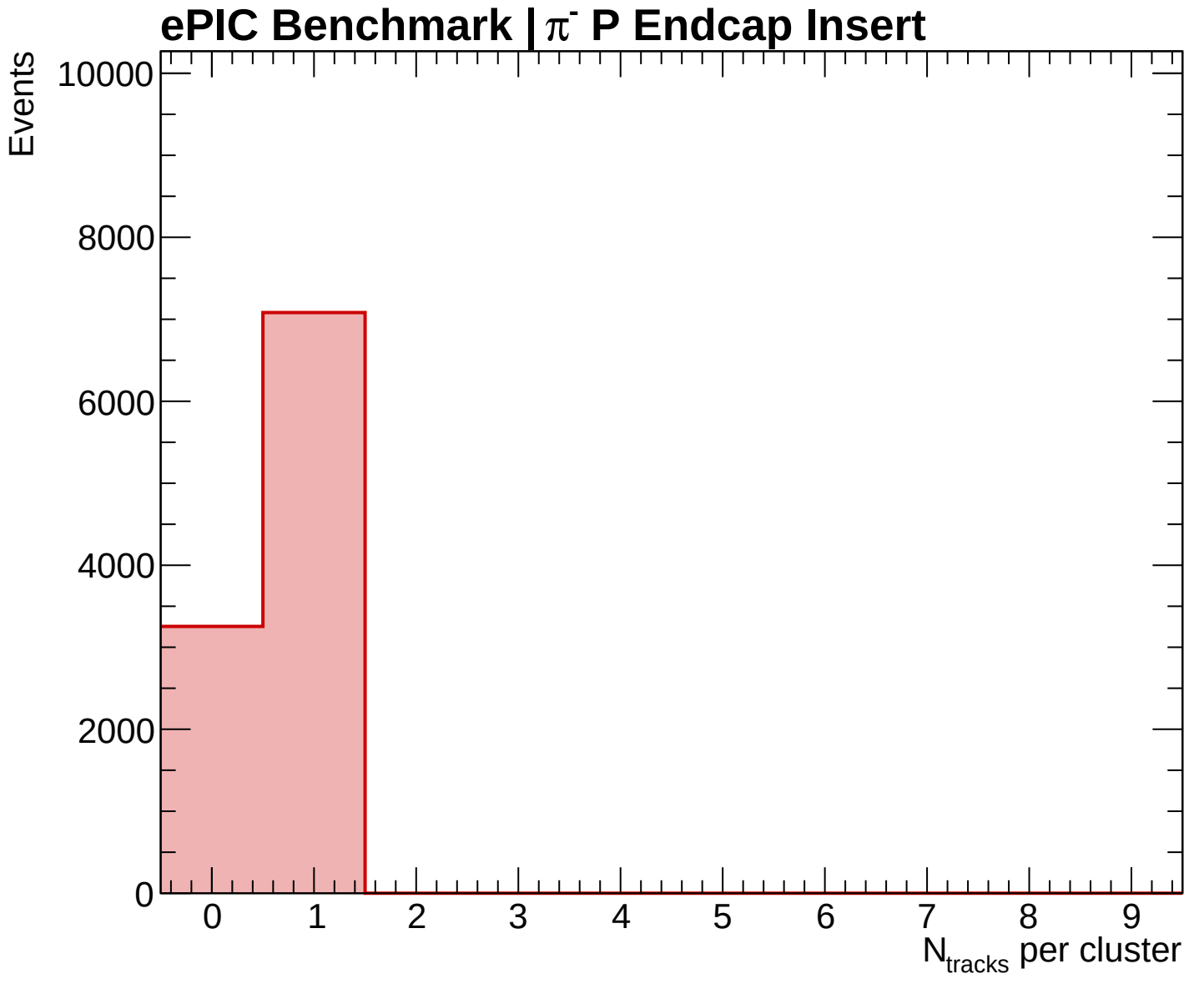

ePIC Benchmark | π⁻ P Endcap Insert
Events
Mean = 0.649
Std = 0.373
Entries = 4123
300
250
200
150
100
50
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_Cl / p_Tr

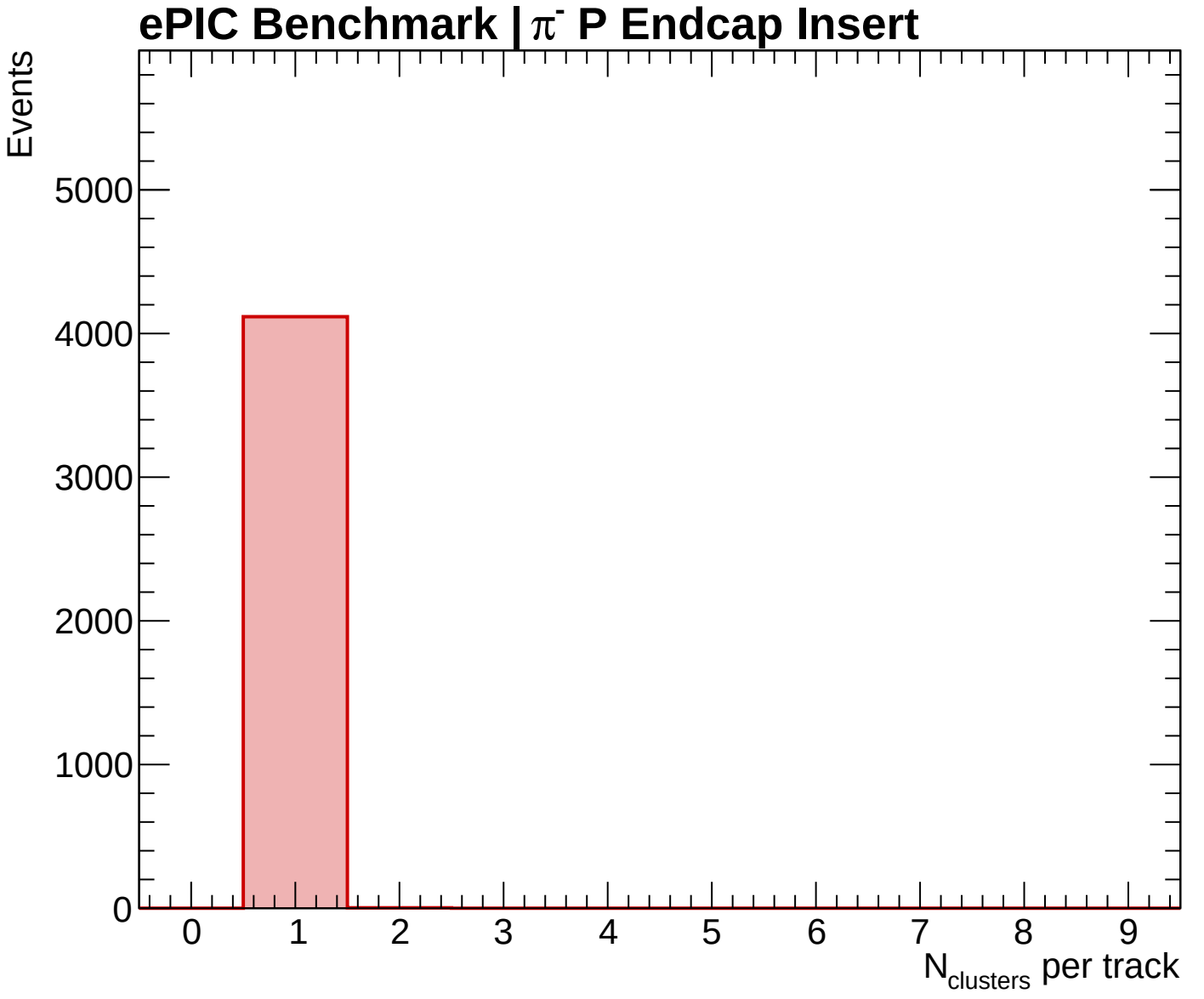

ePIC Benchmark | π⁻ P Endcap Insert
Mean = 0.462
Std = 0.188
Entries = 4123
Events
0
50
100
150
200
250
300
350
400
450
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / E_Cl

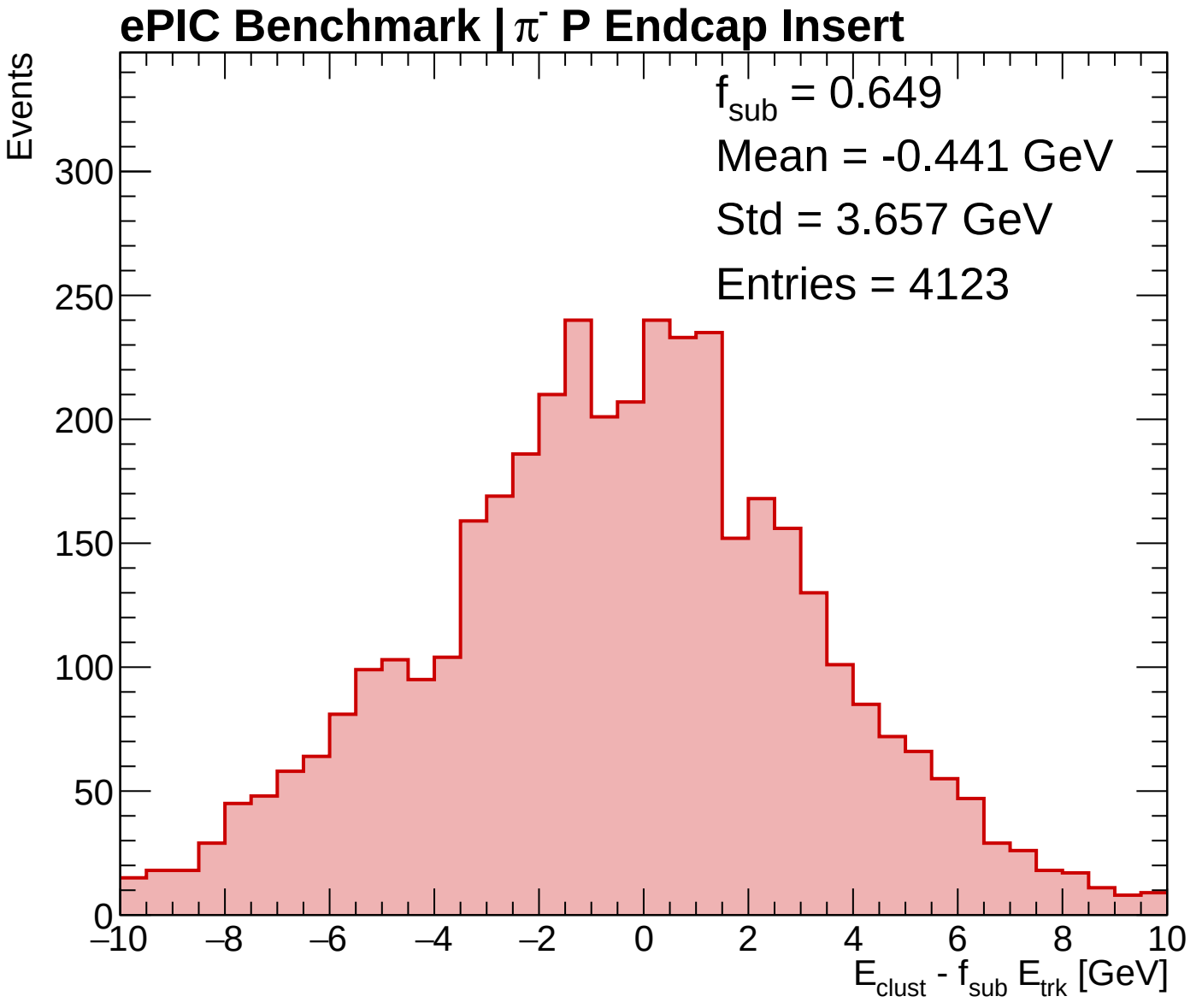

ePIC Benchmark | π⁻ P Endcap Insert
Mean = 0.738
Std = 0.253
Entries = 4123
Events
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / p_Tr


ePIC Benchmark | π⁻ P Endcap Insert
Black: mean profile
f_sub = 0.649
E_Cl / E_Tr
Track p_T [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
2
4
6
8
10
12
14
16


ePIC Benchmark | π⁻ P Endcap Insert
Black: mean profile
f_sub = 0.649
E_Cl / E_Tr
Track η
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
3.1
3.2
3.3
3.4
3.5
3.6
3.7
3.8
18
16
14
12
10
8
6
4
2
0


ePIC Benchmark | π⁻ P Endcap Insert
Events
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N_tracks per cluster


ePIC Benchmark | π- P Endcap Insert
Events
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
N clusters per track


ePIC Benchmark | π⁻ P Endcap Insert
Events
f_sub = 0.649
Mean = -0.441 GeV
Std = 3.657 GeV
Entries = 4123
300
250
200
150
100
50
0
–10
–8
–6
–4
–2
0
2
4
6
8
10
E_clust - f_sub E_trk [GeV]